

the potential of the technology in healthcare applications and their development. LED applications create possibilities for future medical improvements. Some of these are discussed next.

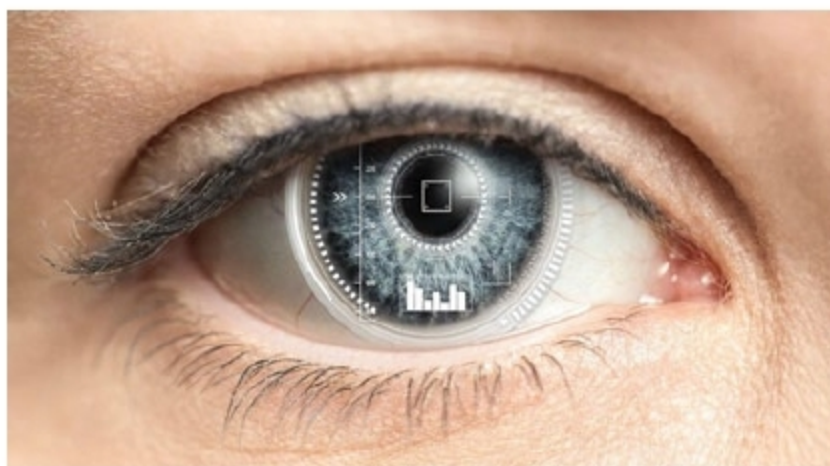
Elastic micro-LED skin sensor system. A new ultra-thin, elastic micro-LED display that fits snugly on the skin has been developed by researchers. The integrated biomedical sensor system, named skin electronics, combines a flexible, deformable display with a lightweight sensor composed of a breathable nano-mesh electrode and wireless communication module. **Flexible skin electronics can measure medical data via the sensor and display information on the screen while transmitting it to the cloud and/or other memory devices.**

The system was developed with the purpose to enhance information accessibility for the elderly or the infirm, who have difficulty operating and obtaining data from existing devices and interfaces.

Micro-LED wearable device to stimulate hair growth. With the development of vertical micro-LEDs, researchers have revealed a new application for re-growing hair for mice. They have developed a wearable photo-stimulator using a flexible array comprising 900 red vertical micro-LEDs. To test the device, they shaved the heads of mice. After twenty days, the mice who received stimulation from the micro-LED device showed longer hair in a wider area in comparison with those with other treatment.



Skin-like sensor can map blood-oxygen levels anywhere in the body (Credit: www.eurekalert.org)



Smart contact lens displays glucose levels in real time (Credit: <https://guardian.ng>)

In terms of its (treatment) advantages, the device used about thousand times less power per unit area as compared to a conventional phototherapeutic laser, without generating heat enough to cause thermal damage to skin.

Smart contact lenses for glucose monitoring and other applications. A Korean research team at Ulsan National Institute of Science and Technology (UNIST) has released smart contact lenses integrated with glucose sensors, wireless power transfer circuits and LEDs. **The device can monitor glucose levels in tears and provide instant sensing results via the embedded LED display.**

In addition, imec has presented its development of LED-light-integrated contact lenses in collaboration with Ghent University and SEED, a contact lens producer. The device has been designed to integrate transducers into lenses for diagnosing and treating ocular diseases. Scientists also suggested that the lenses could provide a wide range of such applications as drug-delivery for eye treatments.

Implantable LED device for curing cancer. **LEDs could also be used to heal cancer, according to researchers from Waseda University.** The Japanese research team has developed an implantable device consisting of LED chips and bio-adhesive nano-sheet, which has successfully shrunk the tumours in mice through photodynamic therapy.

The application can

be attached onto the inner surface of animal tissue where it releases low-intensity irradiation to treat targeted lesions with metronomic photodynamic therapy (mPDT)—a treatment that uses special drugs and particular lights to kill cancer cells with low dose in the long term. By directly attaching the device with mPDT on the target tumor, it reduces the threat to healthy areas.

Flexible OLED sensor to monitor blood-oxygen. A flexible OLED sensor has been developed for detecting blood-oxygen levels of skin, tissue and other organs. The OLED sensor has been assembled from a printed sheet of organic photo detectors on the top and organic red and IR LEDs on the bottom. With organic electronics printed on bendable plastic that moulds to the contours of the body, the sensor can detect blood-oxygen levels at nine points in a grid and can be placed anywhere on the skin. **Through tracking blood-oxygen levels on a larger area, the OLED sensor can monitor healing wounds.** Such disorders as diabetes and respiration diseases could also benefit from the application.

IR pain treatment. A company in the US has received clearance to market its LED-based IR therapy device developed for temporary relief/reduction of minor pain or discomfort associated with muscles including neck and back pain caused by muscle tension, joints, arthritis, sprains and strains. The device makes use of low-intensity IR light from an LED source to stimulate natural biological processes in the area where light is applied. Referred to as photo-bio-modulation or photo-bio-stimulation, the process promotes the body's own natural healing and tissue immune responses.

To conclude, LEDs have already proven their versatility in the healthcare sector. Coloured light, for example, has a calming effect on patients, as dynamically changing the light provides the human organism with activating impetus. This is why hospitals and medical practices are increasingly making use of powerful LEDs for diagnostic and therapeutic purposes. **EFY**